PRD - GoComet Nova Trade Document Pipeline

1. Nova and GoComet Understanding

What is Nova?
Nova is GoComet's agentic AI layer for logistics and trade operations. It is designed to deliver an outcome, not just another dashboard. Instead of as

What is the FDE model?
The Forward Deployed Engineer model means GoComet does not treat implementation as a handoff after a sale. An engineer works closely with the

What is a System of Outcomes?
A System of Outcomes is software that completes work with enough trust and traceability that a user can rely on the result. A System of Record sto

2. Problem Statement

Trade document validation breaks in five predictable places today. Operators read documents line by line, which is slow and inconsistent. Rules ofte

Success in the first five minutes is simple and concrete. A CG operator uploads a document and gets structured output in under 30 seconds. Every

3. Users and Jobs To Be Done

Persona 1 - CG Operator
The CG operator processes 50 to 200 documents each week and is accountable for speed, correctness, and exception handling. Their pain is repeti

Persona 2 - Supplier
The supplier submits shipping paperwork and is usually outside the internal process. Their pain is delayed feedback, vague rejection reasons, and re

JTBD
1. When I receive a new trade document, I want the system to extract key fields automatically so that I do not spend 15 to 20 minutes reading it ma
2. When extraction completes, I want to see a field-by-field pass, fail, or uncertain status so that I know where to focus immediately.
3. When the system is unsure about a field, I want it called out explicitly so that I do not accidentally approve bad data.
4. When a required field mismatches the customer rule set, I want an amendment request drafted automatically so that I can shorten the correction
5. When I review an exception, I want to understand the agent's reasoning so that I can make a faster, defensible decision.
6. When a supplier sends a revised document, I want the pipeline to rerun consistently so that I can confirm the fix without redoing all prior work.

4. Agent Architecture

Three agents is the right boundary for this problem because the steps represent three distinct failure surfaces. The extractor handles perception. Th

The Extractor Agent accepts a PDF or image and returns structured JSON for eight required fields plus confidence per field. The Validator Agent ac

Agents communicate through structured JSON persisted to a shared `pipeline_runs` record keyed by `document_id`. This gives each step a durable

5. LLM and Tooling Choices

For extraction, the best production option is a vision-capable model such as GPT-4o or Claude Sonnet with document understanding because trade

For this POC, the backend uses FastAPI and SQLite because both keep the system portable and easy to demo on a laptop. The orchestrator is a s

6. Trust, Failure Handling, and Evals

Hallucination prevention starts at extraction. Every field includes confidence. If a field is unreadable or below the threshold, it becomes `uncertain` a

The pipeline sets a clear trust policy: uncertainty routes to human review, required mismatches route to an amendment draft, and only strong match

Offline eval: a 20-document set with manually verified truth labels, measuring per-field extraction accuracy and mismatch detection quality. Online m

7. Metrics and Success Criteria

North Star
Percentage of trade documents that reach a correct final decision without human intervention in extraction or validation.

Supporting Metrics
1. Field extraction accuracy by field against ground truth
2. False approval rate on required mismatches
3. Percentage of uncertain fields surfaced instead of silently approved
4. Router decision accuracy against operator-confirmed outcome
5. Average end-to-end latency per document
6. Cost per processed document
7. Human override rate after a system recommendation
8. Supplier amendment acceptance rate

Go / No-Go for a Two-Week Pilot
Go if extraction accuracy exceeds 85 percent on pilot documents, there are zero silent approvals on critical mismatches, latency stays under 45 sec

8. What's Next

The next highest-value step is email ingestion because it removes the manual upload step and makes the workflow feel truly agentic. Second is the